JENNIFER SOBECK

CONTACT

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INTEREST SUMMARY

I am a stellar spectroscopist with an atomic and molecular physics background. My scientific interests lie in the chemodynamical characterization of stellar populations across a range of environments.
In general, I want to apply my varied expertise to large astronomical surveys and work in a collaborative research environment.

EDUCATION

PhD in Physics – University of Texas at Austin 2007
An Atomic Physics Viewpoint of Stellar Abundance Analysis

MA in Physics – University of Texas at Austin 2003
Gas Phase Enthalpy Difference Between the Two Conformers of n-Butane

BS in Physics – University of Texas at Austin

RESEARCH POSITIONS

Lead Systems Scientist Oct 2023 – Present
Roman Science Support Center at Caltech/IPAC
Oversee the Microlensing Science Operations System (MSOS) pipeline development team for the Roman Galactic Bulge Time Domain Survey

Associate Scientist Oct 2023 – Present
IPAC, California Institute of Technology

MSE System Scientist Feb 2021 – Oct 2023
Maunakea Spectroscopic Explorer, Canada France Hawaii Telescope

SDSS-IV Project Manager for APOGEE-2 & LCO Operations Nov 2018 – Feb 2021
Sloan Digital Sky Survey IV

Senior Research Scientist Mar 2017 – Feb 2021
Department of Astronomy, University of Washington

APOGEE-2 Project Manager/Deputy Project Manager Oct 2013 – Nov 2018
Sloan Digital Sky Survey IV

Senior Scientist/Research Associate Oct 2013 – Mar 2017
Department of Astronomy, University of Virginia

External Postdoctoral Scholar Jan 2012 – Dec 2013
Department of Astronomy and Astrophysics, University of Chicago

CNRS Postdoctoral Scholar Dec 2011 – Jan 2013
Laboratoire Lagrange, Observatoire de la Cote d'Azur

Kavli Institute for Cosmological Physics Associate Fellow Jun 2010 – Dec 2012
Department of Astronomy and Astrophysics, University of Chicago

JINA Postdoctoral Scholar Oct 2008 – Dec 2011
Joint Institute for Nuclear Astrophysics
Department of Astronomy and Astrophysics, University of Chicago

DGDF Postdoctoral Fellow 2007, 2008
European Southern Observatory, Garching, Germany

TEACHING POSITIONS

Assistant Instructor 2020, 2022, 2023
Software Carpentry

Guest Lecturer 2018, 2019
Department of Astronomy, University of Washington

Assistant Instructor 2003 – 2005
Dept. Astronomy & Dept. Physics, University of Texas at Austin

Teaching Assistant 2000 – 2003
Department of Physics, University of Texas at Austin

Publications

Overview:

I currently have 214 Astronomy- and Physics-related publications (103 Refereed; 111 Non-Refereed), which have received over 25,000 citations (h-index: 67). NASA ADS Author Query (Link)

Selected Relevant Publications:

[1] Baril, M. R., Sheinis, A. I., Sobeck, J., Frinchaboy, P., Epinat, B., & Ho, K. K., 2024, SPIE Proc., 13096, 130969P9 [doi:10.1117/12.3018448]
"Conceptual Design of an Integral Field Unit for the Canada France Hawaii Telescope"
[2] Sheinis, A., Barden, S.C., & Sobeck, J., 2023, AN, 344, e20230108 [doi:10.1002/asna.20230108]
"The Maunakea Spectroscopic Explorer: Thousands of fibers, infinite possibilities"
[3] Marshall, J. L., Sobeck, J., Petric, A. O., & Flagey, N., 2022, SPIE Proc., 12186, 1218609 [doi:10.1117/12.2629973]
"Planning Scientific Operations for the Maunakea Spectroscopic Explorer"

Talks and Presentations

Overview:

I have given more than 50 presentations on a variety of subjects (science, survey planning/management, teaching, instrumentation, software/coding, and outreach) in multiple forums over my career.

Committees

Roman Core Community Survey Definition Committee: Galactic Bulge Time Domain Survey
Habitable Worlds Observatory Stars & Stellar Populations Working Group Steering Committee
Rubin LSST Survey Cadence Optimization Committee Milky Way Observing Strategy Task Force
Rubin LSST Stars, Milky Way & Local Volume (SMWLV) MC Working Group, Co-Chair

Review Panels

NSF AAG Proposal Review Panelist
NSF AAPF Proposal Review Panelist
NSF CAREER Review Panelist
NASA APRA-SAT Proposal Review Panelist and Panel Chair

Service

STScI HST Proposal Review Panelist and Vice Chair
NOIRLab Gemini Large Program Review Panelist
Astrophysical Journal Referee
Astrophysical Journal Letters Referee
Astronomy & Astrophysics Referee
Journal of Physics B: Atomic, Molecular & Optical Physics Referee
Monthly Notices of the Royal Astronomical Society Referee
United States Congressional Visits Day, Science, Engineering, & Technology

Grants and Awards

SDSS-IV Architect	2020
e-Science Institute Incubator Project (PI)	2018
NSF Astronomy and Astrophysics Research Grants (Collab/Co-I)	AST 0607708, 0907732, 1715898
Sloan Foundation Grant (Collaborator/Co-I)	2013, 2015
University of Virginia Center for Global Inquiry & Innovation Grant (Co-I)	2015
STScI Hubble Space Telescope Approved GO/AR Program Grants (Co-I)	Cycles 18, 19, 20, 23

Skills

Programming	*Proficient:* Python, IDL, Fortran *Familiar:* R, Matlab, Mathematica, C++
Software	*Proficient:* LaTeX, Unix, Git *Familiar:* SQL, Solidworks
Languages	English *(fluent)*, French *(proficient)*